

June 8, 2026 04:01 AM GMT

Semiconductors | North America

# Weekly: NVDA memory cuts? And Apr SIA

Potential reduction in Vera Rubin rack content is another indication of the intensity of the memory shortage; separately, we are raising our industry estimates post a strong April SIA report Friday.

**NVDA DRAM cuts reflect tight supply conditions.** There were reports this week from Semi Analysis that suggested NVIDIA was reducing LPDDR5 memory content from the Vera Rubin racks from 55 TB to 28 TB, reducing the SOCAMM memory module size from 192 GB to 96 GB.

While we are not sure that it will impact every rack, we were able to confirm that there have been some changes and that it least some racks will be shipped with this lower configuration. Given the importance of memory we suspect that the higher configurations will be available, if not immediately, then shortly after shipments begin in F3q.

The rack memory is a significant portion of overall global demand for DRAM; assuming that 53-70k racks are built next year, at 55 TB, the rack SOCAMM would approach 5% of global demand, so cutting that in half across the board - the most draconian figure - would be 2%+ impact (1.4 mm TB reduction for a 62mm TB market), but one which impacts a higher value portion of the market.

That said, all of our contacts would indicate that NVIDIA, and for that matter everyone in cloud, will buy every GB of SOCAMM that they can get, and our assumption would be that if they can catch up then they would return to higher configurations. This is entirely to ensure that the DRAM shortage has minimized impact to GPU rack sales, and is actually evidence that there is a true shortage not a double ordering window - and that incremental supply will be met wit incremental demand.

**Overall SIA data was stronger than expected for April, driven by both broad markets and memory.** April Semiconductor Industry Association billings data reported on Friday, June 5th, came in higher than our estimates and seasonality for broad markets and memory:

- **Overall:** Sales were down 2.2% m/m, above our estimate of -12.1% and above the 10-yr average change of -10.6%. 3-month y/y growth accelerated to 93.9% from 79.1%, and one month y/y growth was 106.4%.
- **Trend by geography (y/y):** The Americas (+158.5%) was followed by Asia Pacific (+113.0%), China (+76.7%), Europe (+64.1%), and Japan (+25.7%).

Broad markets accelerated, following a broadly strong March:

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| Industry View  | Attractive |

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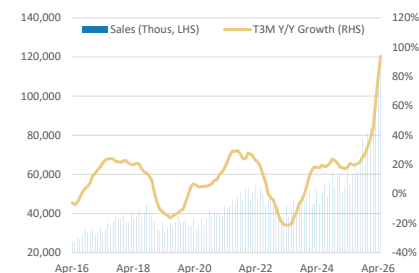
**For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.**

- **Discrete** (beat): -3.4% m/m vs our estimate of -8.0% and the 10-yr average change of -9.7%. Units were above the 10-yr average (0.2% vs -7.2%) while ASP was below (-3.6% vs -2.6%).
- **Analog** (beat): -2.7% m/m vs our estimate of -6.0% and 10-yr average of -9.1%. Units were above the 10-yr average (-7.1% vs -8.9%) and ASP was above (4.4% vs -3.5%).
- **MCU** (beat): -3.1% m/m vs our estimate of -9.0% and 10-yr average change of -12.2%. Units were above the 10-yr average (-7.1% vs -8.9%) and ASP was above (4.4% vs -3.5%).
- **MPU** (beat): -0.2% m/m vs our estimate of -2.5% and 10-yr average change of -6.3%.

April built on an accelerating [March](#), continuing to align with positive JunQ guides from suppliers. From the commentary we have heard intra-quarter, the cycle continues to improve, with Industrial in particular re-accelerating from cyclical lows and broadening out to non-AI-related areas. And while we have seen pricing increases across the supply chain, we'd note that the largest driver is cost pass-through versus value recapture, ex a few companies such as ADI. AI continues to outpace, and while we still are hearing there is no replenishment cycle, visibility into a 2H acceleration looks stronger.

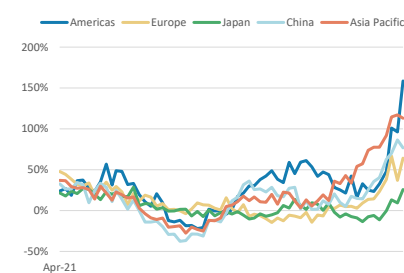
In analog, the 3-month average y/y growth accelerated from 21.9% to 22.4%, led by Application Specific. MCUs' 3-month average y/y growth decelerated to 13.2% from 16.9% in March, with growth decelerating in both General Purpose and Application Specific, though the latter at a slower rate.

**Exhibit 1:** Global Semiconductor Sales



Source: SIA, Morgan Stanley Research

**Exhibit 2:** Semiconductor Sales by Region



Source: SIA, Morgan Stanley Research

**Memory was strong in April**, with both NAND and DRAM outperforming seasonality and our estimates:

- **DRAM:** Above expectation and 5-yr average, coming in at -3.7% m/m vs our estimate of -24.2% m/m and 5-yr average of -27.6%. Bits were above our estimate (-16.5% vs -31.7%) m/m, up 52.8% y/y), while ASP was above (15.3% vs 10.0%). On a 3-month y/y basis, total DRAM sales reached 298.5% in April, marking a new historical high since 2001, while ASP growth (188.1%) has now increased for nine consecutive quarters.

- **NAND:** Above expectation and the 5-yr average, coming in at -4.2% m/m vs our estimate of -11.2% and 5-yr average of -13.9%. Bits were above our estimate (-20.9% m/m vs -22.8%) again likely reflecting constraints, while ASPs were above (up 21.0% m/m vs 15.0%- prices were up 281.6% y/y ). On a 3-month average y/y basis, revenue reached 307.0% in April, also marking a new record in our dataset's history, while ASP growth of 213.4% also marked a new record and volume decelerated slightly to 30.7% from 36.8% in March.

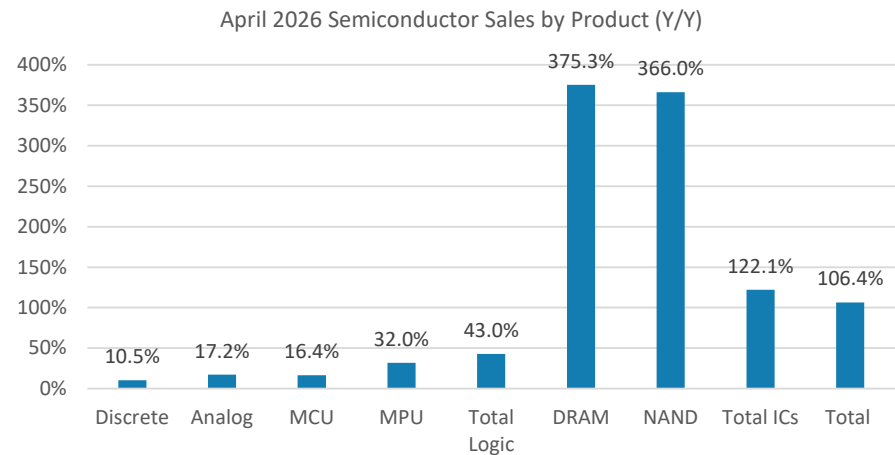
The April data reinforces our view that there is no quick fix to the memory shortage, with DRAM increasingly the principal bottleneck to the AI buildout and NAND also very tight as hyperscalers continue to absorb higher-performance supply. We think the strength is still primarily supply/demand driven rather than an inventory-led cycle, with HBM wafer intensity, cleanroom/EUV constraints, and limited incremental NAND capacity all supporting a higher-for-longer pricing environment. While investors remain focused on the durability debate, we would view LTAs as a symptom of tight supply and hyperscaler urgency to secure capacity, rather than the cause of the cycle. We continue to view memory supply as a critical constraint on AI, which should support unusually strong DRAM pricing, a more durable NAND recovery, and continued ownership of that exposure through MU and SNDK.

**Changes to our forecast:** Our forecast comes up from +91% to +103% for this year, reflecting strength across the board but mainly memory, where our CY26 forecast increases from \$807bn to \$880bn, as 3q pricing is starting to look higher than our previous forecast. For CY27, we are assuming 22% growth y/y to \$1.96 trillion, mostly due to memory pricing rippling through.

**Our take:** April SIA adds another datapoint that this is less of a narrow AI cycle and increasingly a broader supply-constrained upcycle. Memory remains the most obvious expression of that, with DRAM now a principal bottleneck to the AI buildout and NAND also tightening as hyperscalers absorb higher-performance supply. We would continue to own that exposure through MU and SNDK, particularly given the lack of a quick supply response and the increasing evidence that pricing strength can last longer than investors had expected. At the same time, the better-than-seasonal broad-market data reinforces what suppliers have been saying intra-quarter: Industrial is re-accelerating from cyclical lows, demand is broadening into non-AI areas, and inventory digestion is moving further into the rear-view. That combination keeps us constructive on leading-edge logic beneficiaries including NVDA, AVGO, and ALAB, cap equipment / supply-chain beneficiaries including LRCX, KLAC, and MKSI, and broader analog / MCU suppliers such as ADI, NXP, and ALGM as the broad-market cycle accelerates.

SIA Note Charts

Exhibit 3: April 2026 Semiconductor Sales by Product (Y/Y)



Source: SIA, Morgan Stanley Research

Exhibit 4: Variance Table

| Reported Item (\$ mn)           | April 2026 Actual | April 2026 Est. | Difference (Act-Est) | Last Mth    | MoM    | Last Yr     | YoY    | Last Mth (Before Revision) | Last Mth Revised vs. Reported |
|---------------------------------|-------------------|-----------------|----------------------|-------------|--------|-------------|--------|----------------------------|-------------------------------|
| Discrete / Opto / Sensors Sales | 8,521             | 8,117           | 5.0%                 | 8,823       | -3.4%  | 7,714       | 10.5%  | 8,823                      |                               |
| Analog Sales                    | 7,917             | 7,650           | 3.5%                 | 8,138       | -2.7%  | 6,757       | 17.2%  | 8,138                      |                               |
| MCU Sales                       | 2,049             | 1,924           | 6.5%                 | 2,114       | -3.1%  | 1,760       | 16.4%  | 2,114                      | 0                             |
| MPU Sales                       | 5,745             | 5,614           | 2.3%                 | 5,758       | -0.2%  | 4,352       | 32.0%  | 5,758                      | 0                             |
| Total Micro Sales               | 8,017             | 7,754           | 3.4%                 | 8,088       | -0.9%  | 6,361       | 26.0%  | 8,088                      | 0                             |
| Total Logic (ex Micro) Sales    | 32,569            | 31,417          | 3.7%                 | 32,388      | 0.6%   | 22,026      | 47.9%  | 32,388                     |                               |
| Total Logic Sales               | 40,586            | 39,171          | 3.6%                 | 40,476      | 0.3%   | 28,387      | 43.0%  | 40,476                     |                               |
| DRAM Sales                      | 37,566            | 29,573          | 27.0%                | 39,028      | -3.7%  | 7,904       | 375.3% | 39,028                     | 0                             |
| DRAM Gigabit Equivalents        | 28,417,085        | 23,447,461      | 21.2%                | 34,037,628  | -16.5% | 18,598,712  | 52.8%  | 34,037,628                 | 0                             |
| DRAM Price per Gb Equivalent    | \$1.3220          | \$1.2613        | 4.8%                 | \$1.1466    | 15.3%  | \$0.4250    | 211.1% | \$1.1466                   | \$0.0000                      |
| NAND Sales                      | 18,349            | 17,016          | 7.8%                 | 19,160      | -4.2%  | 3,937       | 366.0% | 19,160                     |                               |
| NAND 1 Gigabit Equivalent       | 613,243,576       | 598,396,859     | 2.5%                 | 774,855,320 | -20.9% | 502,098,632 | 22.1%  | 774,855,320                |                               |
| NAND Price per Gb Equivalent    | \$0.0299          | \$0.0284        | 5.2%                 | \$0.0247    | 21.0%  | \$0.0078    | 281.6% | \$0.0247                   |                               |
| Total Memory Sales              | 56,654            | 47,174          | 20.1%                | 58,772      | -3.6%  | 12,208      | 364.1% | 58,772                     | 0                             |
| Total ICs                       | 105,156           | 93,994          | 11.9%                | 107,386     | -2.1%  | 47,351      | 122.1% | 107,386                    | 0                             |
| Semiconductor Sales             | 113,677           | 102,112         | 11.3%                | 116,210     | -2.2%  | 55,066      | 106.4% | 116,210                    | 0                             |

Source: SIA, Morgan Stanley Research

Exhibit 5: Quarterly SIA Data

|                             | Mar/24  | Jun/24  | Sep/24  | Dec/24  | Mar/25  | Jun/25  | Sep/25  | Dec/25  | Mar/26  | Jun/26  | Sep/26  | Dec/26  |
|-----------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Revenues (\$ Millions)      | A       | A       | A       | A       | A       | A       | A       | A       | A       | E       | E       | E       |
| Discretes / Optos / Sensors | 22,185  | 21,383  | 24,285  | 23,191  | 21,519  | 22,629  | 25,449  | 24,905  | 23,996  | 26,151  | 28,532  | 28,368  |
| Analog                      | 19,276  | 19,011  | 20,648  | 20,653  | 19,813  | 20,186  | 23,052  | 23,396  | 22,756  | 24,209  | 26,095  | 26,468  |
| MCU                         | 5,751   | 5,409   | 5,512   | 5,087   | 4,950   | 5,326   | 5,694   | 5,628   | 5,788   | 6,466   | 6,735   | 6,722   |
| MPU                         | 12,124  | 13,276  | 13,911  | 14,860  | 13,606  | 13,943  | 15,894  | 17,615  | 16,548  | 18,078  | 19,410  | 20,579  |
| Other                       | 651     | 658     | 690     | 703     | 728     | 752     | 813     | 788     | 565     | 669     | 100     | 100     |
| Total Micro                 | 18,526  | 19,343  | 20,114  | 20,650  | 19,284  | 20,021  | 22,401  | 24,011  | 22,902  | 25,213  | 26,245  | 27,400  |
| Logic (ex Micro)            | 49,452  | 49,479  | 56,194  | 61,634  | 65,356  | 68,892  | 78,853  | 88,783  | 91,117  | 101,338 | 109,583 | 117,024 |
| Total Logic                 | 66,988  | 68,822  | 76,307  | 82,283  | 84,640  | 88,913  | 101,254 | 112,794 | 114,019 | 126,550 | 135,828 | 144,424 |
| DRAM                        | 18,175  | 22,269  | 26,173  | 28,243  | 27,074  | 31,635  | 39,949  | 51,939  | 93,436  | 147,692 | 163,222 | 177,551 |
| NAND                        | 13,472  | 17,813  | 18,003  | 17,140  | 12,617  | 15,404  | 17,382  | 22,282  | 42,794  | 76,274  | 84,702  | 89,738  |
| Other                       | 1,039   | 1,010   | 1,133   | 1,046   | 1,070   | 1,154   | 1,310   | 1,339   | 1,544   | 2,214   | 2,214   | 2,214   |
| Total Memory                | 32,685  | 41,092  | 45,308  | 46,431  | 40,761  | 48,193  | 58,641  | 75,550  | 137,775 | 226,179 | 250,138 | 269,503 |
| Total ICs                   | 116,949 | 128,925 | 142,263 | 149,367 | 145,214 | 157,293 | 182,946 | 211,740 | 274,550 | 376,939 | 412,061 | 440,396 |
| Total Semiconductor         | 141,134 | 150,308 | 166,547 | 172,558 | 166,732 | 179,921 | 208,395 | 236,645 | 298,546 | 403,090 | 440,593 | 468,783 |
| Q/Q Change                  | -3.3%   | 6.5%    | 10.8%   | 3.6%    | -3.4%   | 7.9%    | 15.8%   | 13.6%   | 26.2%   | 35.0%   | 9.3%    | 6.4%    |
| Y/Y Change                  | 18.1%   | 16.6%   | 23.7%   | 18.2%   | 18.1%   | 19.7%   | 25.1%   | 37.1%   | 79.1%   | 124.0%  | 111.4%  | 98.1%   |

Source: SIA, Morgan Stanley Research

## Exhibit 6: Annual SIA Data

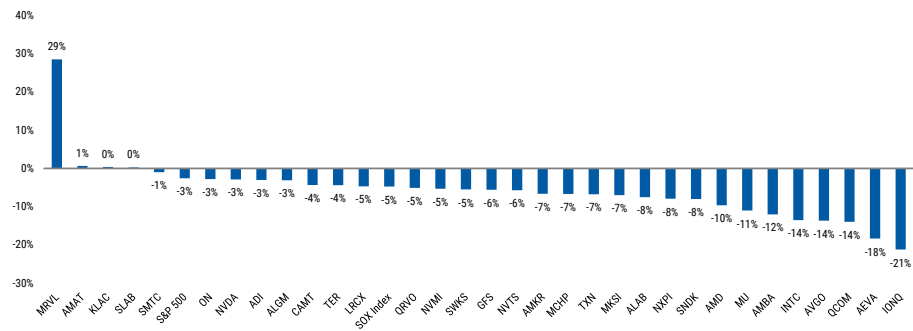
|                               | 2010           | 2017           | 2018           | 2019           | 2020           | 2021           | 2022           | 2023           | 2024           | 2025           | 2026             | 2027             |
|-------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|------------------|
|                               | A              | A              | A              | A              | A              | A              | A              | A              | A              | A              | E                | E                |
| <b>Revenues (\$ Millions)</b> |                |                |                |                |                |                |                |                |                |                |                  |                  |
| Discretes / Optos / Sensors   | 48,406         | 69,087         | 75,446         | 78,953         | 79,164         | 92,945         | 99,591         | 98,359         | 91,044         | 94,502         | 107,066          | 114,712          |
| Analog                        | 42,285         | 53,069         | 58,380         | 53,939         | 55,658         | 73,816         | 88,919         | 81,146         | 79,588         | 86,447         | 99,529           | 108,504          |
| MCU                           | 14,799         | 16,430         | 17,027         | 15,808         | 15,484         | 19,622         | 25,030         | 27,861         | 21,758         | 21,597         | 25,711           | 28,489           |
| MPU                           | 39,927         | 44,370         | 46,371         | 47,974         | 51,812         | 56,766         | 51,299         | 45,480         | 54,172         | 61,059         | 74,614           | 83,018           |
| Other                         | 5,908          | 3,286          | 3,262          | 2,658          | 2,382          | 2,845          | 3,217          | 3,172          | 2,702          | 3,062          | 1,434            | 400              |
| <b>Total Micro</b>            | <b>60,633</b>  | <b>64,086</b>  | <b>66,660</b>  | <b>66,440</b>  | <b>69,678</b>  | <b>79,234</b>  | <b>79,546</b>  | <b>76,513</b>  | <b>78,633</b>  | <b>85,718</b>  | <b>101,760</b>   | <b>111,907</b>   |
| Logic (ex Micro)              | 77,377         | 102,196        | 109,411        | 106,535        | 118,408        | 153,710        | 176,097        | 178,493        | 215,768        | 301,883        | 419,062          | 500,028          |
| <b>Total Logic</b>            | <b>138,010</b> | <b>166,281</b> | <b>176,071</b> | <b>172,975</b> | <b>188,086</b> | <b>232,944</b> | <b>255,643</b> | <b>255,006</b> | <b>294,401</b> | <b>387,601</b> | <b>520,822</b>   | <b>611,935</b>   |
| DRAM                          | 39,210         | 72,802         | 98,604         | 62,475         | 64,324         | 92,960         | 77,769         | 51,945         | 94,860         | 150,598        | 581,901          | 675,059          |
| NAND                          | 21,735         | 47,227         | 54,263         | 40,180         | 49,390         | 55,953         | 47,109         | 36,104         | 66,427         | 67,685         | 293,508          | 445,443          |
| Other                         | 8,669          | 3,946          | 4,428          | 3,785          | 3,769          | 4,921          | 4,948          | 4,261          | 4,229          | 4,229          | 4,229            | 4,229            |
| <b>Total Memory</b>           | <b>69,614</b>  | <b>123,976</b> | <b>157,295</b> | <b>106,440</b> | <b>117,482</b> | <b>153,834</b> | <b>129,825</b> | <b>92,309</b>  | <b>165,516</b> | <b>222,512</b> | <b>879,638</b>   | <b>1,124,732</b> |
| <b>Total ICs</b>              | <b>249,909</b> | <b>343,326</b> | <b>391,746</b> | <b>333,354</b> | <b>361,226</b> | <b>460,594</b> | <b>474,388</b> | <b>428,461</b> | <b>539,505</b> | <b>696,560</b> | <b>1,499,989</b> | <b>1,845,171</b> |
| <b>Total Semiconductor</b>    | <b>298,315</b> | <b>412,414</b> | <b>467,192</b> | <b>412,307</b> | <b>440,389</b> | <b>553,540</b> | <b>573,978</b> | <b>526,820</b> | <b>630,549</b> | <b>791,061</b> | <b>1,607,055</b> | <b>1,959,883</b> |
| Y/Y % Growth                  | 31.8%          | 21.7%          | 13.3%          | -11.7%         | 6.8%           | 25.7%          | 3.7%           | -8.2%          | 19.7%          | 25.5%          | 103.2%           | 22.0%            |
| 5 year CAGR                   | 6.0%           | 7.2%           | 8.9%           | 4.2%           | 5.6%           | 10.3%          | 6.8%           | 2.4%           | 8.5%           | 12.4%          | 23.8%            | 27.8%            |
| 10 year CAGR                  | 3.9%           | 4.9%           | 6.5%           | 6.2%           | 4.0%           | 6.3%           | 7.0%           | 5.6%           | 6.5%           | 9.0%           | 16.8%            | 16.9%            |
| Y/Y Real Growth               | 30.0%          | 19.5%          | 11.0%          | -13.1%         | 6.0%           | 21.7%          | -3.4%          | -12.3%         | 16.7%          | 23.0%          | 99.2%            | 19.6%            |

Source: SIA, Morgan Stanley Research



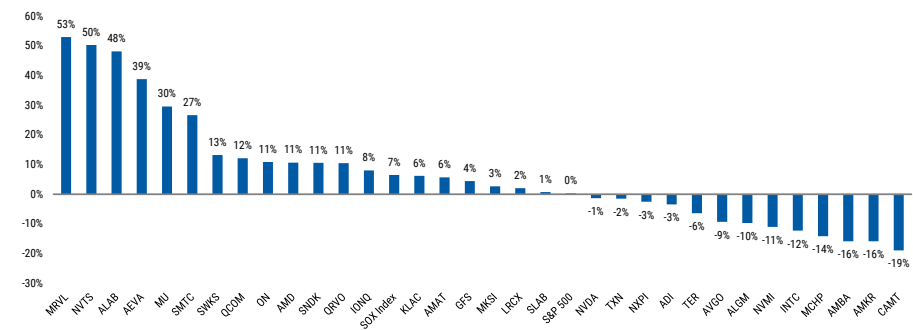
# Semiconductor Coverage – Valuation / Stock Performance

Exhibit 7: Weekly stock performance



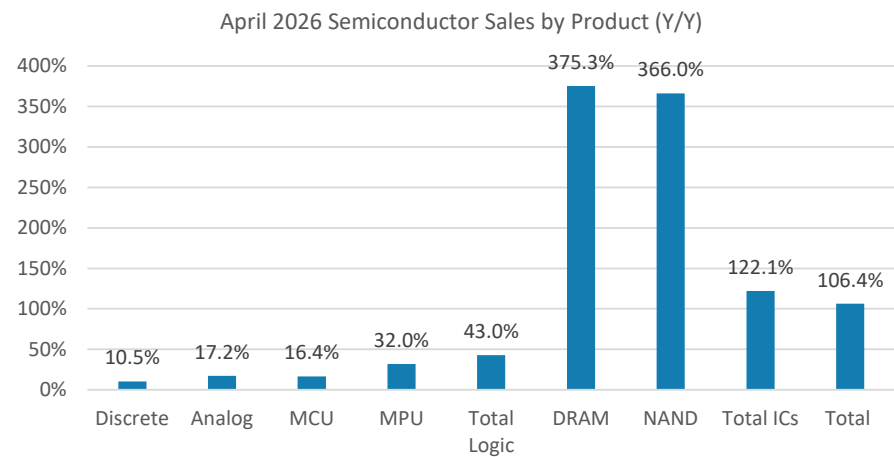
Source: FactSet, Morgan Stanley Research

Exhibit 8: Monthly stock performance



Source: FactSet, Morgan Stanley Research

Exhibit 9: YTD stock performance



Source: FactSet, Morgan Stanley Research

# Semis Week Ahead

**Exhibit 10:** June 8th - June 12th, 2026

**Monday, June 8th**

**Tuesday, June 9th**

**PCIM Europe -Day 1**

<https://pcim.mesago.com/nuernberg/en/expo.html>

**Wednesday, June 10th**

**PCIM Europe -Day 2**

<https://pcim.mesago.com/nuernberg/en/expo.html>

**Thursday, June 11th**

**PCIM Europe -Day 3**

<https://pcim.mesago.com/nuernberg/en/expo.html>

**Friday, June 12th**

**LOOKING AHEAD**

**Wednesday, June 24th**

**QCOM Investor Day**

**MU Earnings**

4:30pm ET

**Thursday, June 25th**

**Renesas Capital Markets Day**

**Thursday, July 23rd**

**AMD Advancing AI**

<https://www.youtube.com/@AMD/streams>

**Wednesday, September 16th**

**ON Financial Analyst Day**

2:00pm ET

Source: Bloomberg, Refinitiv, Morgan Stanley Research



# Semiconductor Coverage

**Exhibit 11:** Semiconductors - Risk Reward

| Ticker                               | Rating | Price     | Risk-Reward |         |         | % Upside<br>/ Downside |
|--------------------------------------|--------|-----------|-------------|---------|---------|------------------------|
|                                      |        |           | Bear        | Base    | Bull    | Base                   |
| <b><u>Computing</u></b>              |        |           |             |         |         |                        |
| AMD                                  | EW     | \$466.4   | \$210       | \$410   | \$570   | -12%                   |
| INTC                                 | EW     | \$99.2    | \$43        | \$73    | \$110   | -26%                   |
| NVDA                                 | OW     | \$205.1   | \$160       | \$288   | \$330   | 40%                    |
| <b><u>Memory</u></b>                 |        |           |             |         |         |                        |
| MU                                   | OW     | \$864.0   | \$675       | \$1,050 | \$1,650 | 22%                    |
| SNDK                                 | OW     | \$1,559.3 | \$1,100     | \$1,750 | \$2,635 | 12%                    |
| <b><u>Capital Equipment</u></b>      |        |           |             |         |         |                        |
| AMAT                                 | EW     | \$453.0   | \$330       | \$502   | \$693   | 11%                    |
| CAMT                                 | EW     | \$164.2   | \$118       | \$163   | \$214   | -1%                    |
| KLAC                                 | OW     | \$1,929.2 | \$1,367     | \$1,900 | \$2,334 | -2%                    |
| LRCX                                 | OW     | \$303.3   | \$219       | \$331   | \$436   | 9%                     |
| MKSI                                 | OW     | \$301.7   | \$234       | \$374   | \$506   | 24%                    |
| NVMI                                 | EW     | \$475.8   | \$366       | \$494   | \$643   | 4%                     |
| TER                                  | EW     | \$357.9   | \$267       | \$387   | \$512   | 8%                     |
| <b><u>Comm-IC / Multi-Market</u></b> |        |           |             |         |         |                        |
| ALAB                                 | OW     | \$317.1   | \$121       | \$240   | \$332   | -24%                   |
| AMBA                                 | OW     | \$63.5    | \$34        | \$96    | \$136   | 51%                    |
| AVGO                                 | OW     | \$385.7   | \$308       | \$502   | \$637   | 30%                    |
| MRVL                                 | EW     | \$263.5   | \$130       | \$195   | \$286   | -26%                   |
| QCOM                                 | UW     | \$215.9   | \$94        | \$146   | \$199   | -32%                   |
| QRVO                                 | EW     | \$98.3    | \$40        | \$84    | \$110   | -15%                   |
| SMTC                                 | EW     | \$151.0   | \$95        | \$175   | \$230   | 16%                    |
| SWKS                                 | EW     | \$73.6    | \$43        | \$76    | \$105   | 3%                     |
| <b><u>Analog / MCUs</u></b>          |        |           |             |         |         |                        |
| ADI                                  | OW     | \$401.4   | \$342       | \$428   | \$517   | 7%                     |
| ALGM                                 | OW     | \$46.4    | \$43        | \$51    | \$70    | 10%                    |
| MCHP                                 | EW     | \$88.3    | \$69        | \$94    | \$118   | 6%                     |
| NVTS                                 | UW     | \$25.1    | \$4.9       | \$13.7  | \$22.7  | -45%                   |
| NXPI                                 | OW     | \$296.0   | \$268       | \$335   | \$400   | 13%                    |
| ON                                   | EW     | \$117.3   | \$62        | \$87    | \$115   | -26%                   |
| SLAB                                 | EW     | \$218.1   | \$83        | \$231   | \$265   | 6%                     |
| TXN                                  | UW     | \$285.1   | \$184       | \$221   | \$263   | -22%                   |
| <b><u>Components</u></b>             |        |           |             |         |         |                        |
| AEVA                                 | EW     | \$23.0    | \$4.5       | \$18.5  | \$52.0  | -20%                   |
| <b><u>Foundry</u></b>                |        |           |             |         |         |                        |
| GFS                                  | EW     | \$75.5    | \$34        | \$58    | \$96    | -23%                   |
| AMKR                                 | EW     | \$65.0    | \$49        | \$69    | \$88    | 6%                     |
| <b><u>Quantum Computing</u></b>      |        |           |             |         |         |                        |
| IONQ                                 | EW     | \$56.8    | \$6         | \$48.5  | \$110   | -15%                   |

Source: FactSet, Morgan Stanley Research estimates

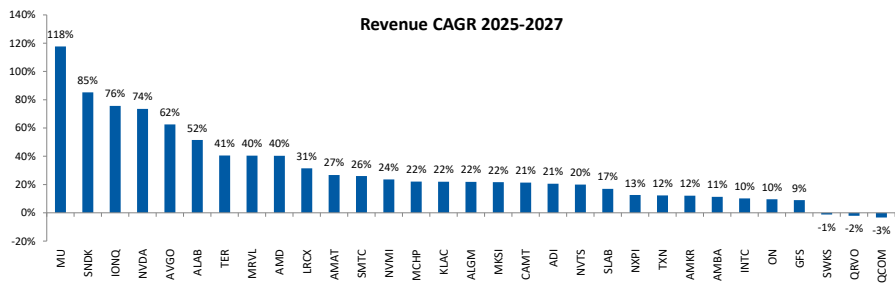
Exhibit 12: MS vs. Street

|    |      | CY2026e EPS |           |        |      | CY2026e Revenue |           |        |
|----|------|-------------|-----------|--------|------|-----------------|-----------|--------|
|    |      | MS          | Consensus | % Diff |      | MS              | Consensus | % Diff |
| OW | ADI  | \$13.21     | \$12.69   | (4%)   | ADI  | \$15,499        | \$15,069  | (3%)   |
|    | ALAB | \$2.88      | \$2.97    | (8%)   | ALAB | \$1,532         | \$1,547   | (1%)   |
|    | ALGM | \$0.93      | \$0.88    | 10%    | ALGM | \$1,048         | \$1,017   | 2%     |
|    | AMBA | \$0.74      | \$0.80    | 3%     | AMBA | \$443           | \$439     | 0%     |
|    | AVGO | \$13.68     | \$12.73   | (3%)   | AVGO | \$123,943       | \$115,415 | 0%     |
|    | KLAC | \$43.86     | \$43.70   | (1%)   | KLAC | \$15,335        | \$15,318  | (0%)   |
|    | LRCX | \$7.00      | \$6.83    | (0%)   | LRCX | \$27,067        | \$26,926  | (0%)   |
|    | MKSI | \$11.70     | \$11.70   | 4%     | MKSI | \$4,793         | \$4,805   | 1%     |
|    | MU   | \$81.93     | \$73.71   | 45%    | MU   | \$144,942       | \$132,538 | 18%    |
|    | NVDA | \$4.77      | \$8.55    | 6%     | NVDA | \$215,938       | \$374,813 | 5%     |
| EW | NXPI | \$14.96     | \$14.68   | 4%     | NXPI | \$14,138        | \$14,028  | 1%     |
|    | SNDK | \$161.07    | \$125.16  | 43%    | SNDK | \$38,078        | \$32,035  | 7%     |
|    | AEVA | -\$1.80     | -\$1.68   | 2%     | AEVA | \$33            | \$33      | (19%)  |
|    | AMAT | \$13.75     | \$12.90   | 2%     | AMAT | \$36,416        | \$34,815  | 3%     |
|    | AMD  | \$7.22      | \$7.47    | 1%     | AMD  | \$48,807        | \$49,875  | (5%)   |
|    | AMKR | \$2.07      | \$2.08    | (8%)   | AMKR | \$7,570         | \$7,610   | (4%)   |
|    | CAMT | \$3.56      | \$3.50    | 10%    | CAMT | \$577           | \$571     | 5%     |
|    | GFS  | \$1.94      | \$1.89    | (4%)   | GFS  | \$7,250         | \$7,238   | (1%)   |
|    | INTC | \$1.12      | \$1.08    | 21%    | INTC | \$58,131        | \$58,496  | 0%     |
|    | IONQ | -\$0.54     | -\$0.35   | (109%) | IONQ | \$266           | \$263     | 3%     |
|    | MCHP | \$2.71      | \$2.78    | (1%)   | MCHP | \$5,769         | \$5,811   | 2%     |
|    | MRVL | \$3.98      | \$3.93    | 3%     | MRVL | \$11,498        | \$11,188  | 2%     |
|    | NVMI | \$10.12     | \$10.47   | 1%     | NVMI | \$1,057         | \$1,064   | 4%     |
|    | ON   | \$3.12      | \$3.09    | 8%     | ON   | \$6,472         | \$6,466   | (0%)   |
|    | QRVO | \$6.31      | \$6.91    | (3%)   | QRVO | \$3,346         | \$3,514   | (7%)   |
| UW | SLAB | \$3.00      | \$2.77    | 9%     | SLAB | \$934           | \$922     | 1%     |
|    | SMTC | \$2.72      | \$2.58    | (2%)   | SMTC | \$1,375         | \$1,331   | 0%     |
|    | SWKS | \$4.68      | \$5.04    | (4%)   | SWKS | \$3,750         | \$3,947   | (3%)   |
| UW | TER  | \$8.03      | \$7.37    | 0%     | TER  | \$4,748         | \$4,518   | 2%     |
|    | NVTS | -\$0.17     | -\$0.17   | 13%    | NVTS | \$43            | \$42      | 15%    |
|    | QCOM | \$9.39      | \$10.69   | (100%) | QCOM | \$40,152        | \$42,538  | (100%) |
| UW | TXN  | \$7.82      | \$7.60    | (1%)   | TXN  | \$21,035        | \$20,927  | (1%)   |

Source: FactSet, Morgan Stanley Research estimates

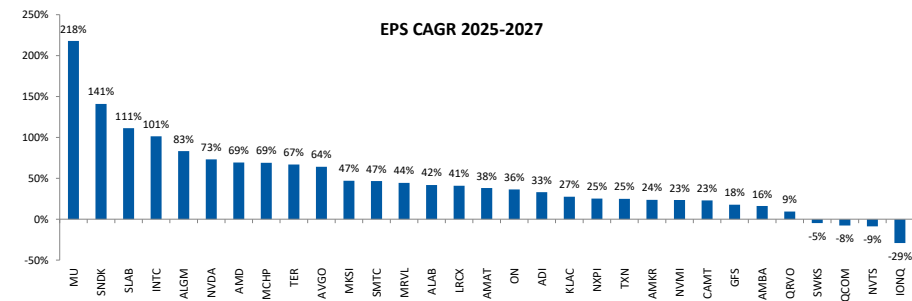
# Revenue and EPS Growth

Exhibit 13: 2025-27e Revenue CAGR



Source: FactSet, Morgan Stanley estimates

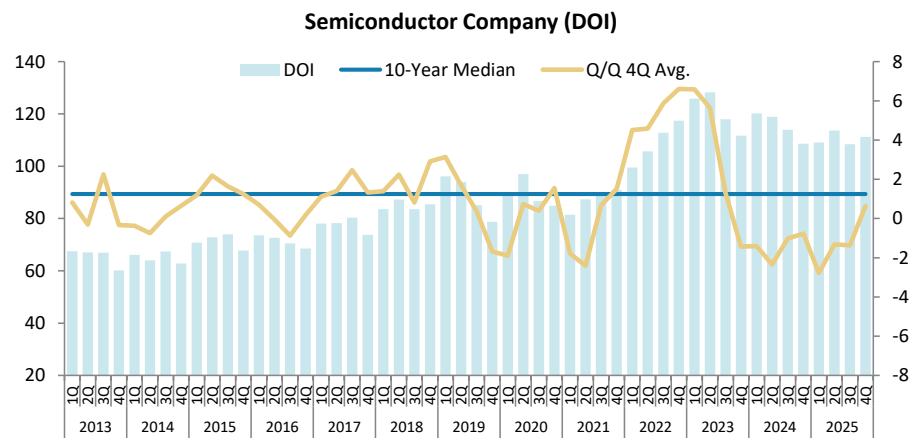
Exhibit 14: 2025-27e EPS CAGR



Source: FactSet, Morgan Stanley estimates

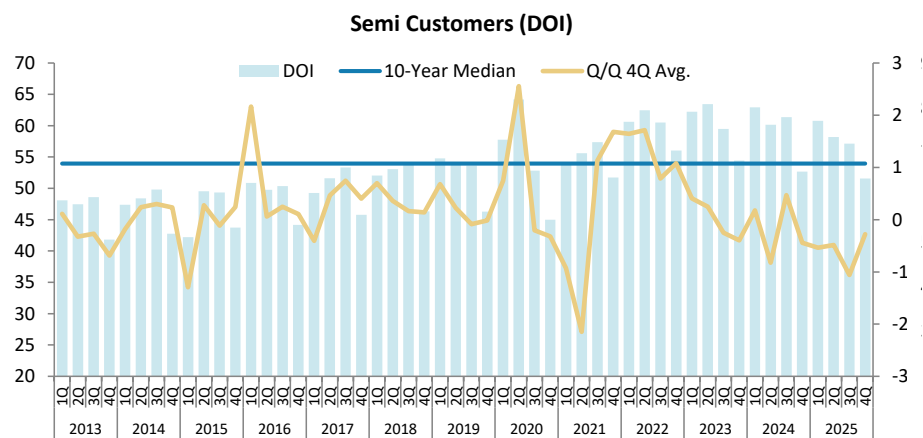
## Inventory

**Exhibit 15:** Semiconductor Company inventory is at 111 days, up 3 days q/q which is 4 days behind of the seasonal decrease of 1 days. DOI is 22 days above the historical median and trailing 4-quarters slightly increased..



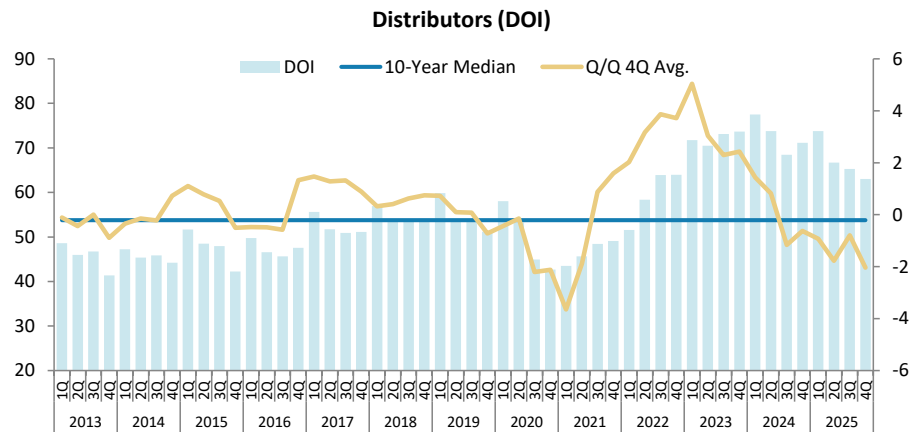
Source: FactSet, Morgan Stanley Research

**Exhibit 16:** Semi Customer DOI decreased 6 days sequentially to 52 days, below the seasonal decrease of 3 days. DOI is below the historical median and trailing 4 quarters increased from last quarter.



Source: FactSet, Morgan Stanley Research

**Exhibit 17:** Distributor inventory is at 63 days, down 2 days q/q, below the seasonal decline of 0 days. DOI is 9 days above the historical median while trailing 4-quarters decreased sequentially.



Source: FactSet, Morgan Stanley Research

## Short Interest

**Exhibit 18:** Short Interest as % of Float

|         | 5/Jun/25 | 6/Mar/26 | 5/Apr/26 | 6/May/26 | 5/Jun/26 |
|---------|----------|----------|----------|----------|----------|
| ADI     | 1.5%     | 1.7%     | 2.0%     | 2.2%     | 1.9%     |
| AEVA    | 5.8%     | 15.0%    | 13.3%    | 12.7%    | 12.5%    |
| ALAB    | 6.2%     | 6.3%     | 9.0%     | 7.7%     | 7.2%     |
| ALGM    | 7.0%     | 5.5%     | 6.4%     | 5.9%     | 5.6%     |
| AMAT    | 2.1%     | 1.7%     | 1.6%     | 2.0%     | 2.3%     |
| AMBA    | 3.4%     | 4.5%     | 4.8%     | 6.0%     | 5.5%     |
| AMD     | 2.4%     | 2.1%     | 2.2%     | 2.2%     | 2.7%     |
| AMKR    | 2.7%     | 3.4%     | 3.5%     | 3.0%     | 3.0%     |
| AVGO    | 1.0%     | 1.0%     | 1.1%     | 1.1%     | 1.1%     |
| CAMT    | 6.8%     | 6.8%     | 7.6%     | 6.6%     | 6.4%     |
| GFS     | 1.7%     | 1.9%     | 3.1%     | 1.8%     | 1.7%     |
| INTC    | 2.1%     | 2.4%     | 2.4%     | 2.8%     | 2.8%     |
| IONQ    | 9.6%     | 23.0%    | 21.7%    | 22.4%    | 19.6%    |
| KLAC    | 2.7%     | 2.3%     | 2.4%     | 2.4%     | 2.6%     |
| LRCX    | 2.0%     | 2.1%     | 2.2%     | 2.3%     | 2.6%     |
| MCHP    | 5.6%     | 4.0%     | 5.1%     | 5.5%     | 5.4%     |
| MKSI    | 5.7%     | 5.7%     | 5.5%     | 5.3%     | 6.4%     |
| MRVL    | 2.7%     | 3.7%     | 3.2%     | 3.2%     | 3.7%     |
| MU      | 2.8%     | 2.6%     | 2.8%     | 3.3%     | 3.1%     |
| NVDA    | 0.9%     | 1.0%     | 1.2%     | 1.2%     | 1.2%     |
| NVMI    | 5.0%     | 4.2%     | 5.2%     | 4.8%     | 5.4%     |
| NVTS    | 12.2%    | 18.8%    | 18.6%    | 17.9%    | 16.5%    |
| NXPI    | 2.3%     | 3.6%     | 3.2%     | 2.6%     | 3.1%     |
| ON      | 9.4%     | 8.2%     | 7.8%     | 7.4%     | 9.0%     |
| QCOM    | 2.0%     | 3.6%     | 4.1%     | 4.7%     | 4.8%     |
| QRVO    | 6.7%     | 7.6%     | 7.1%     | 6.7%     | 7.1%     |
| SLAB    | 5.6%     | 5.1%     | 4.6%     | 5.7%     | 7.2%     |
| SMTC    | 7.7%     | 5.0%     | 5.8%     | 6.7%     | 5.8%     |
| SNDK    | 4.3%     | 5.8%     | 5.4%     | 7.3%     | 6.2%     |
| SWKS    | 9.3%     | 14.3%    | 13.3%    | 15.1%    | 16.3%    |
| TER     | 4.1%     | 2.9%     | 3.1%     | 3.5%     | 3.9%     |
| TXN     | 2.1%     | 2.1%     | 2.0%     | 1.7%     | 2.0%     |
| AVERAGE | 4.6%     | 5.6%     | 5.7%     | 5.8%     | 5.8%     |
| MEDIAN  | 4.1%     | 4.0%     | 4.6%     | 4.8%     | 5.4%     |

Source: FactSet, Morgan Stanley Research



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|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
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| Equal-weight/Hold     | 1571              | 43%        | 369                              | 40%            | 23%                  | 723                                               | 44%                   |
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| Underweight/Sell      | 551               | 15%        | 86                               | 9%             | 16%                  | 201                                               | 12%                   |
| Total                 | 3,667             |            | 920                              |                |                      | 1632                                              |                       |

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## INDUSTRY COVERAGE: Semiconductors

| COMPANY (TICKER)                      | RATING (AS OF) | PRICE* (06/05/2026) |
|---------------------------------------|----------------|---------------------|
| <b>Joseph Moore</b>                   |                |                     |
| Advanced Micro Devices (AMD.O)        | E (06/09/2024) | \$466.38            |
| Aeva Technologies Inc (AEVA.O)        | E (07/19/2021) | \$23.01             |
| Allegro Microsystems Inc (ALGM.O)     | O (02/13/2026) | \$46.39             |
| Ambarella Inc (AMBA.O)                | O (03/29/2016) | \$63.52             |
| Amkor Technology Inc (AMKR.O)         | E (11/08/2023) | \$64.95             |
| Analog Devices Inc. (ADI.O)           | O (11/16/2023) | \$401.39            |
| Astera Labs Inc (ALAB.O)              | O (05/11/2025) | \$317.06            |
| Broadcom Inc. (AVGO.O)                | O (06/09/2024) | \$385.73            |
| GlobalFoundries Inc (GFS.O)           | E (10/28/2024) | \$75.53             |
| Intel Corporation (INTC.O)            | E (02/22/2023) | \$99.17             |
| IonQ Inc (IONQ.N)                     | E (04/25/2023) | \$56.78             |
| Marvell Technology Group Ltd (MRVL.O) | E (09/14/2015) | \$263.47            |
| Microchip Technology Inc. (MCHP.O)    | E (07/10/2024) | \$88.34             |
| Micron Technology Inc. (MU.O)         | O (10/06/2025) | \$864.01            |
| Navitas Semiconductor Corp (NVT.S.O)  | U (04/06/2025) | \$25.08             |
| NVIDIA Corp. (NVDA.O)                 | O (03/16/2023) | \$205.10            |
| NXP Semiconductor NV (NXPI.O)         | O (02/11/2025) | \$295.96            |
| ON Semiconductor Corp. (ON.O)         | E (05/11/2025) | \$117.26            |

|                                     |                 |            |
|-------------------------------------|-----------------|------------|
| Qorvo Inc (QRVO.O)                  | E (10/28/2025)  | \$98.28    |
| Qualcomm Inc. (QCOM.O)              | U (02/10/2026)  | \$215.94   |
| SanDisk Corporation. (SNDK.O)       | O (03/03/2025)  | \$1,559.32 |
| Semtech Corp. (SMTC.O)              | E (04/06/2025)  | \$151.02   |
| Silicon Laboratories Inc. (SLAB.O)  | E (01/19/2021)  | \$218.11   |
| Skyworks Solutions Inc (SWKS.O)     | E (11/28/2018)  | \$73.57    |
| Texas Instruments (TXN.O)           | U (04/13/2020)  | \$285.06   |
| Wolfspeed, INC (WOLF.N)             | NR (04/06/2025) | \$55.06    |
| <b>Lee Simpson</b>                  |                 |            |
| Arm Holdings plc (ARM.O)            | E (04/07/2026)  | \$342.93   |
| Cadence Design Systems Inc (CDNS.O) | O (02/14/2024)  | \$376.19   |
| Synopsys Inc. (SNPS.O)              | E (02/27/2026)  | \$464.85   |

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\* Historical prices are not split adjusted.